Weekly Report

Dean Reiter

April 28, 2026

Readout Timing

Synchron plugin

- Uses variables from GEM data
  - “mpdtime”
    - units of 1/25 ns?
  - “ppgtime”
  - values stored for each GEM
- From TDC data
  - “coarse” = tdc signal time
  - Can loop back to zero if it reaches maximum?
  - Units of 1/10 ns?
  - “chan_coarse” = time of first tdc signal

TrigScint plugin

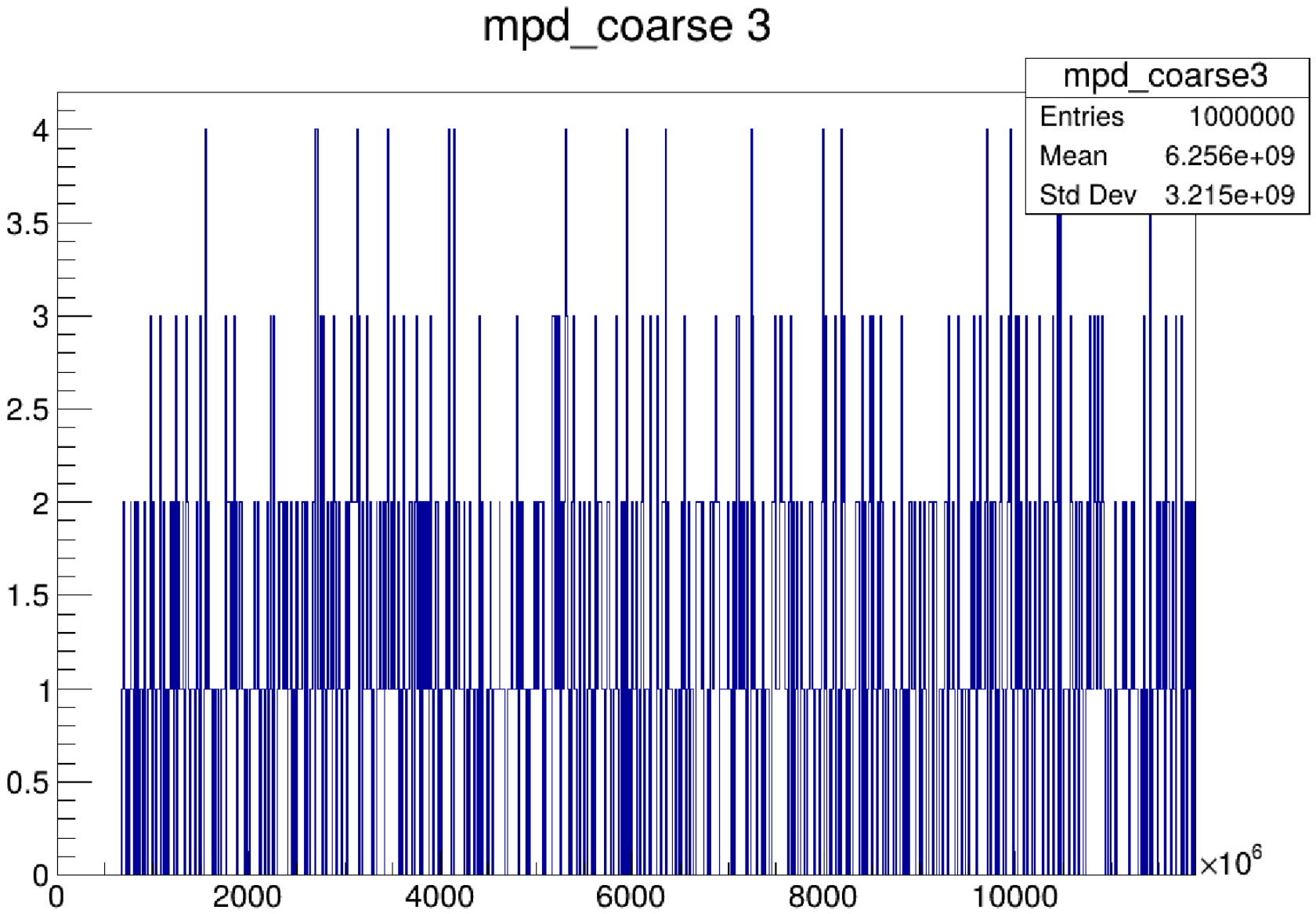
- TDC “hit” time = modified and scaled “coarse” with fine time and offset calibrations

Data

- Run 3932
  - Cosmic
  - Right+Trigger
  - 1M events
  - 9845 seconds (2 hours 44 minutes 5 seconds)
    - 10^{13} nanoseconds

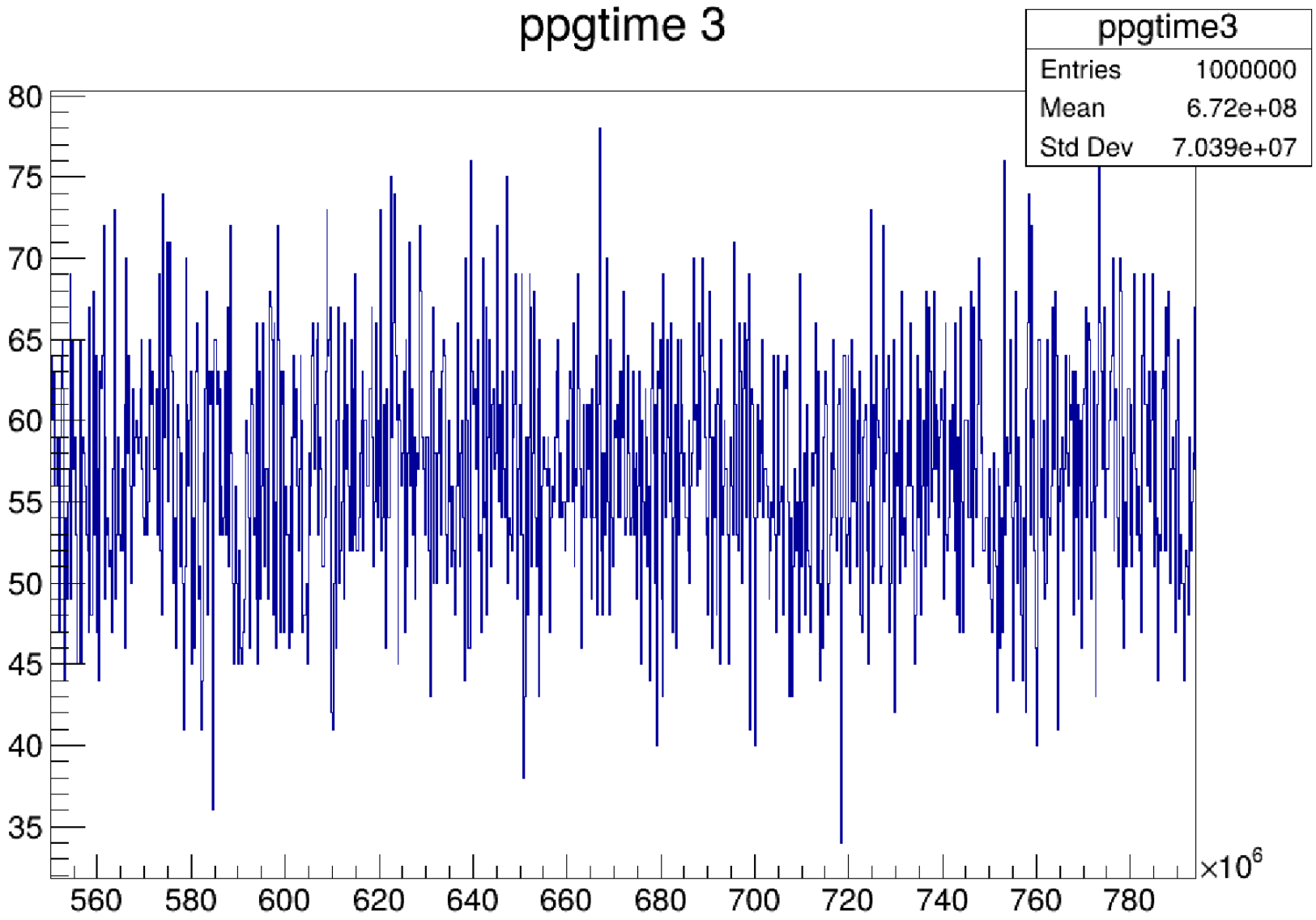
MPD time

- scaled by 25
- Right top GEM
- Range: 0 to 10^{10}
- microseconds?



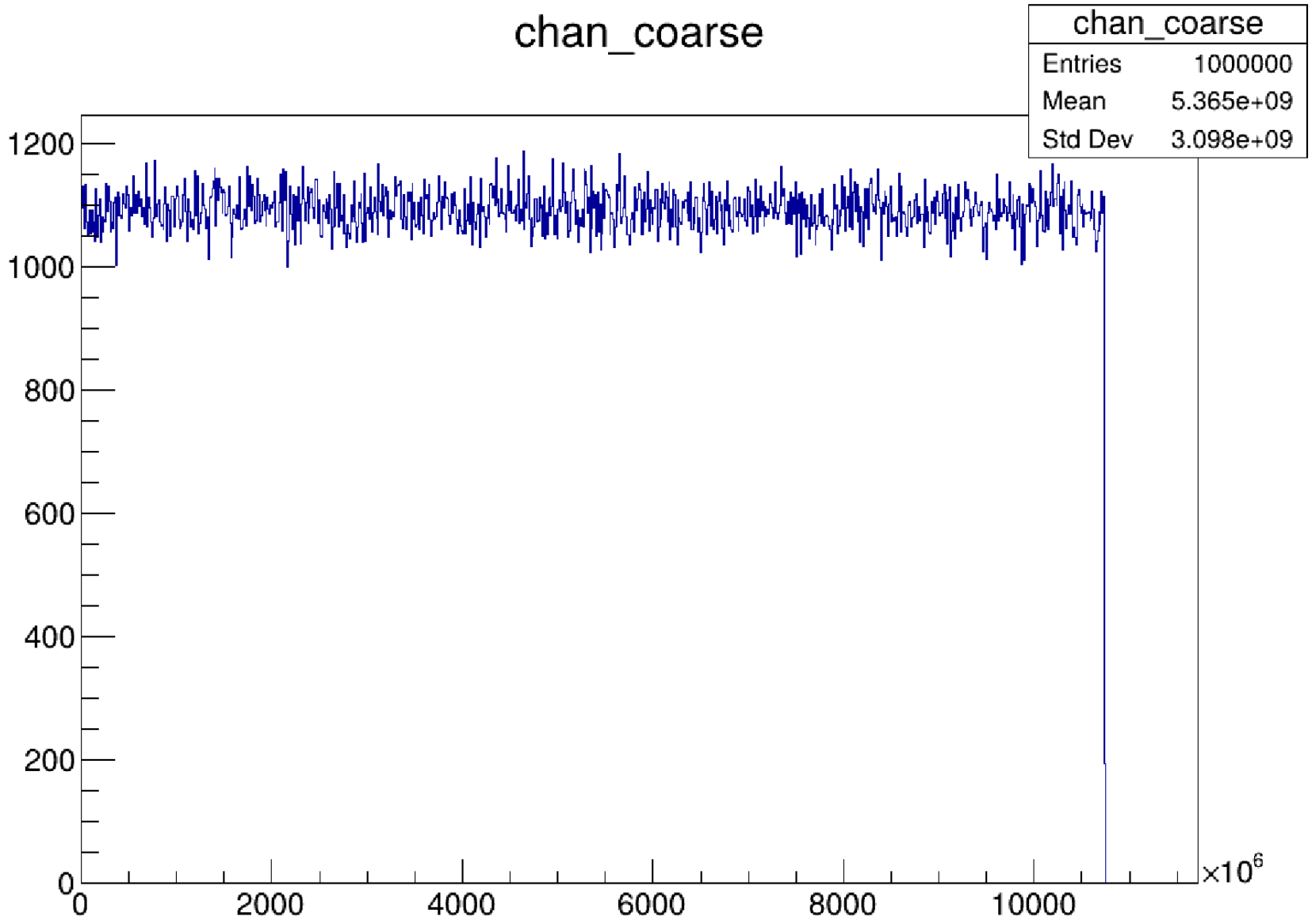
PPG time

- not scaled
- Right top GEM
- Range: 0 to 10^9
  - If you scale by 25, then range is 0 to 2 x 10^{10}
- Unknown units

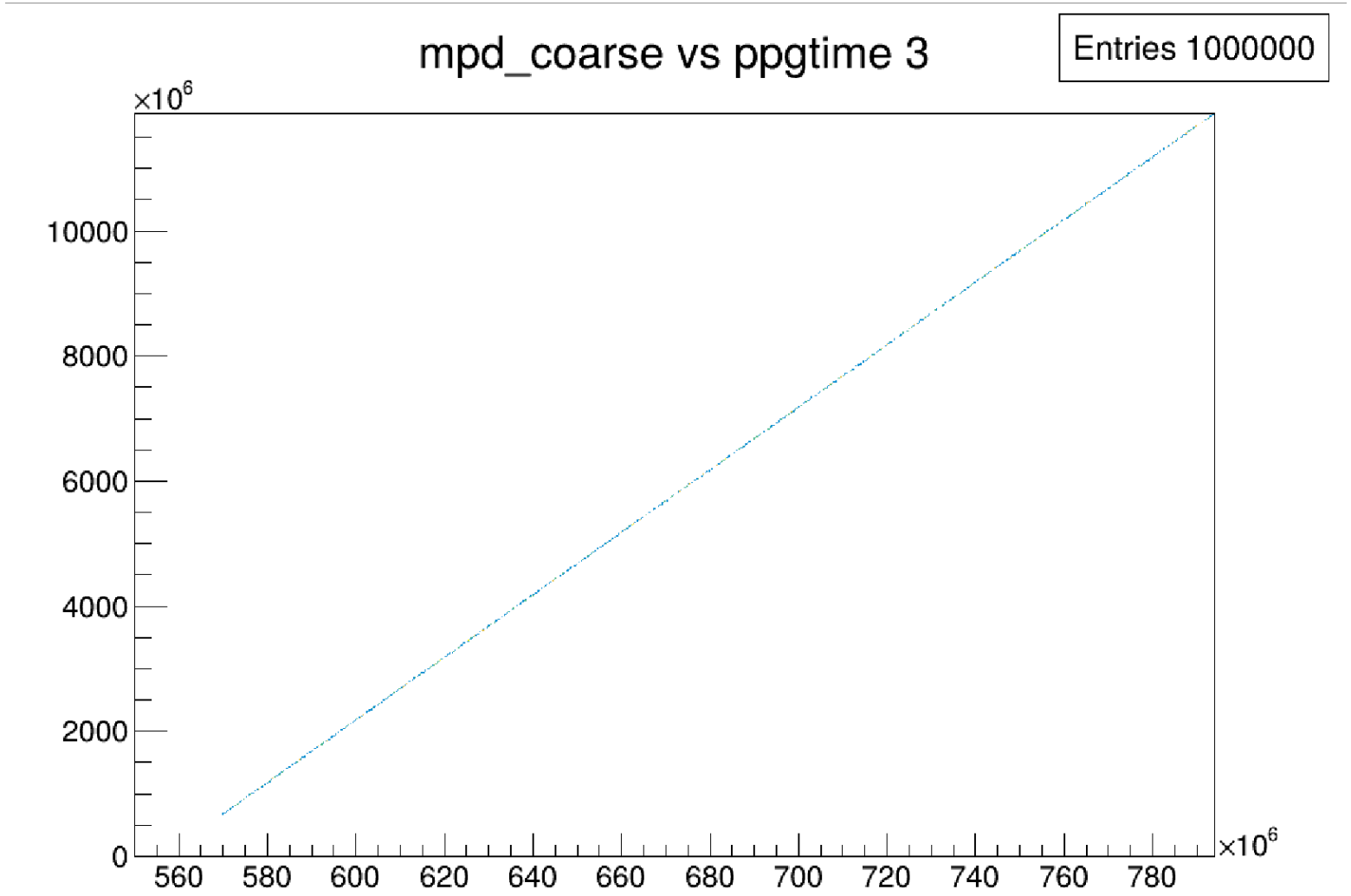


TDC time

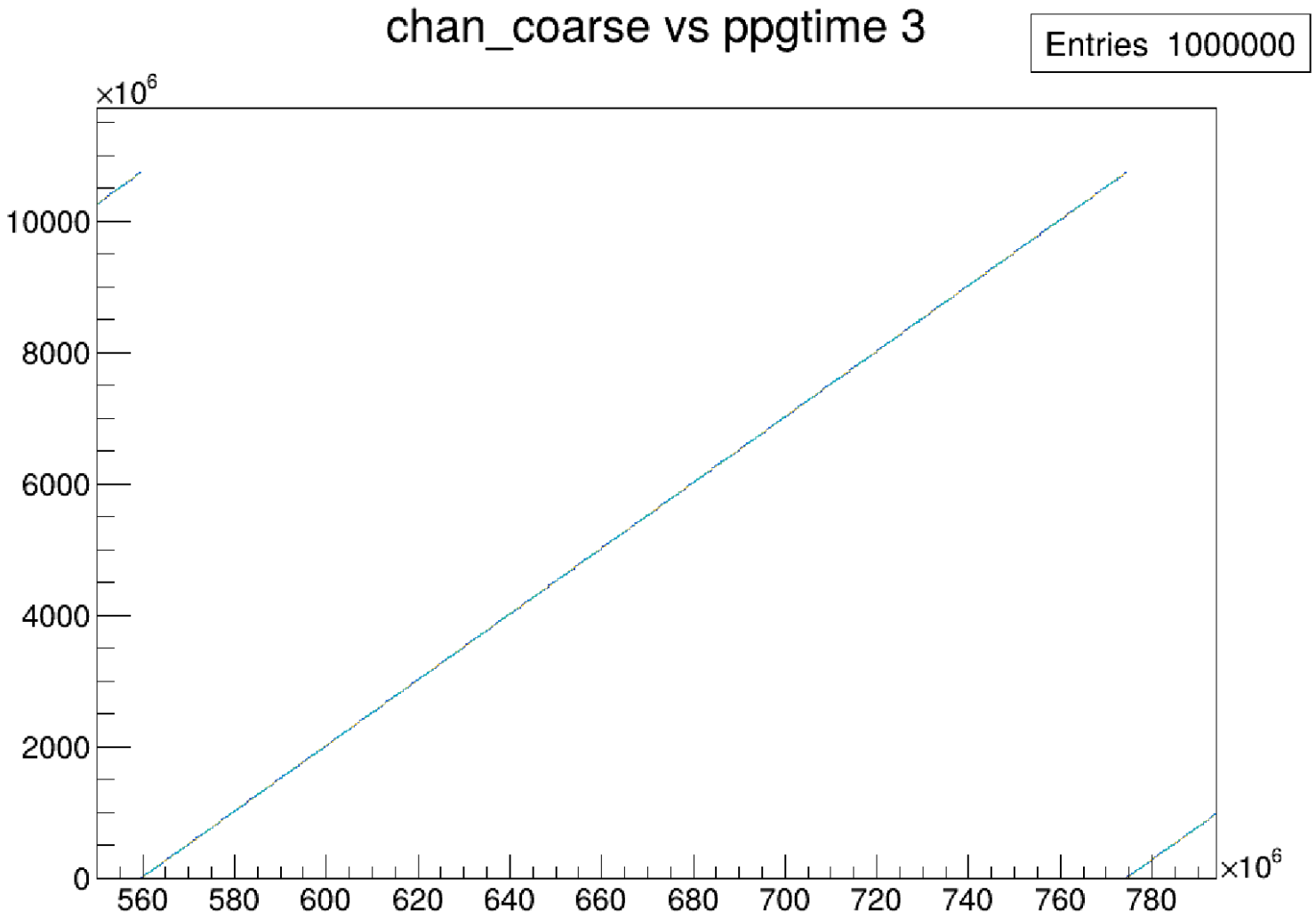
- scaled by 10
- First (lowest) TDC time per event
- Range: 0 to 10^{10}
- Same units as MPD time?



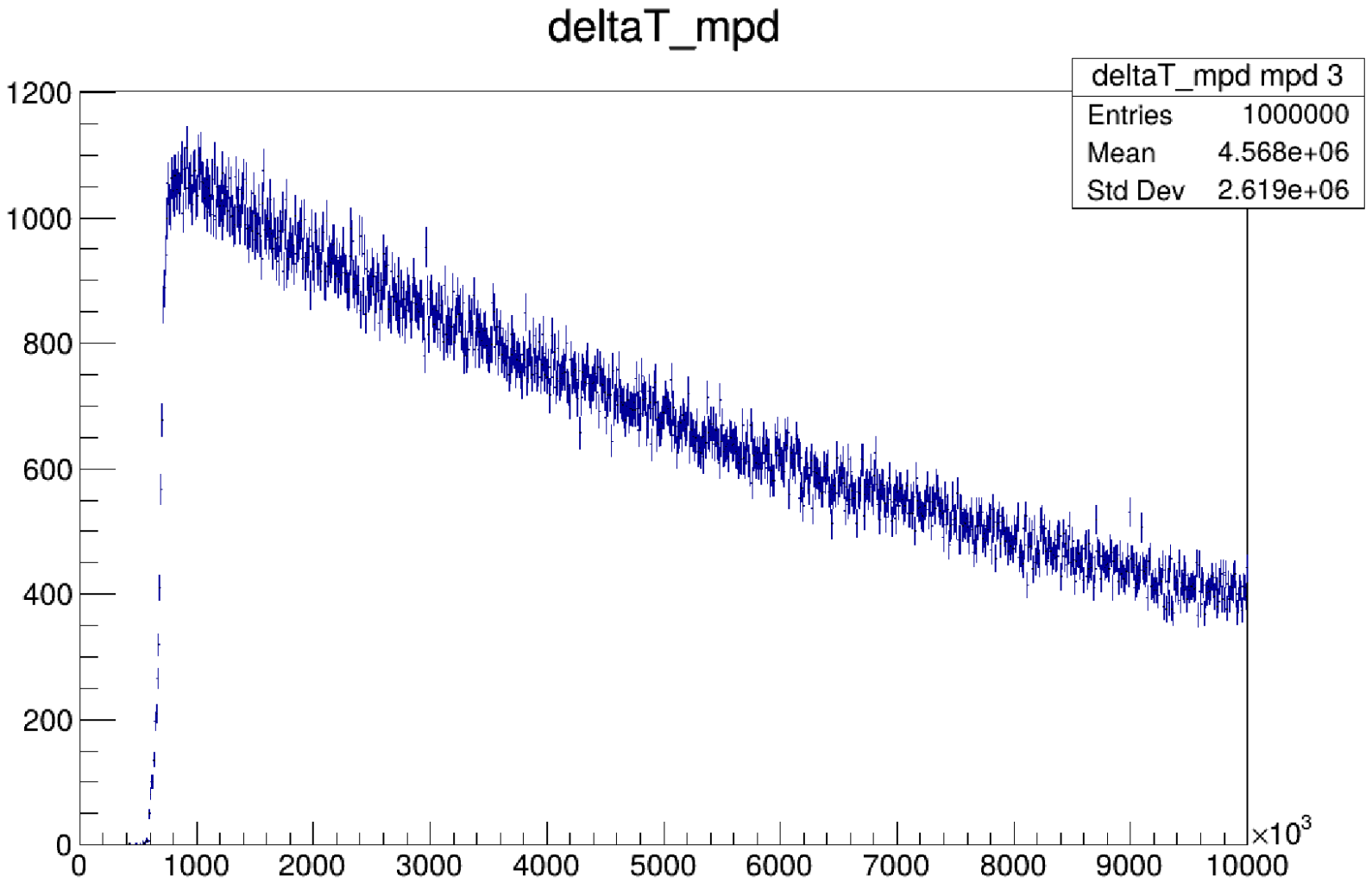
MPD time vs PPG time



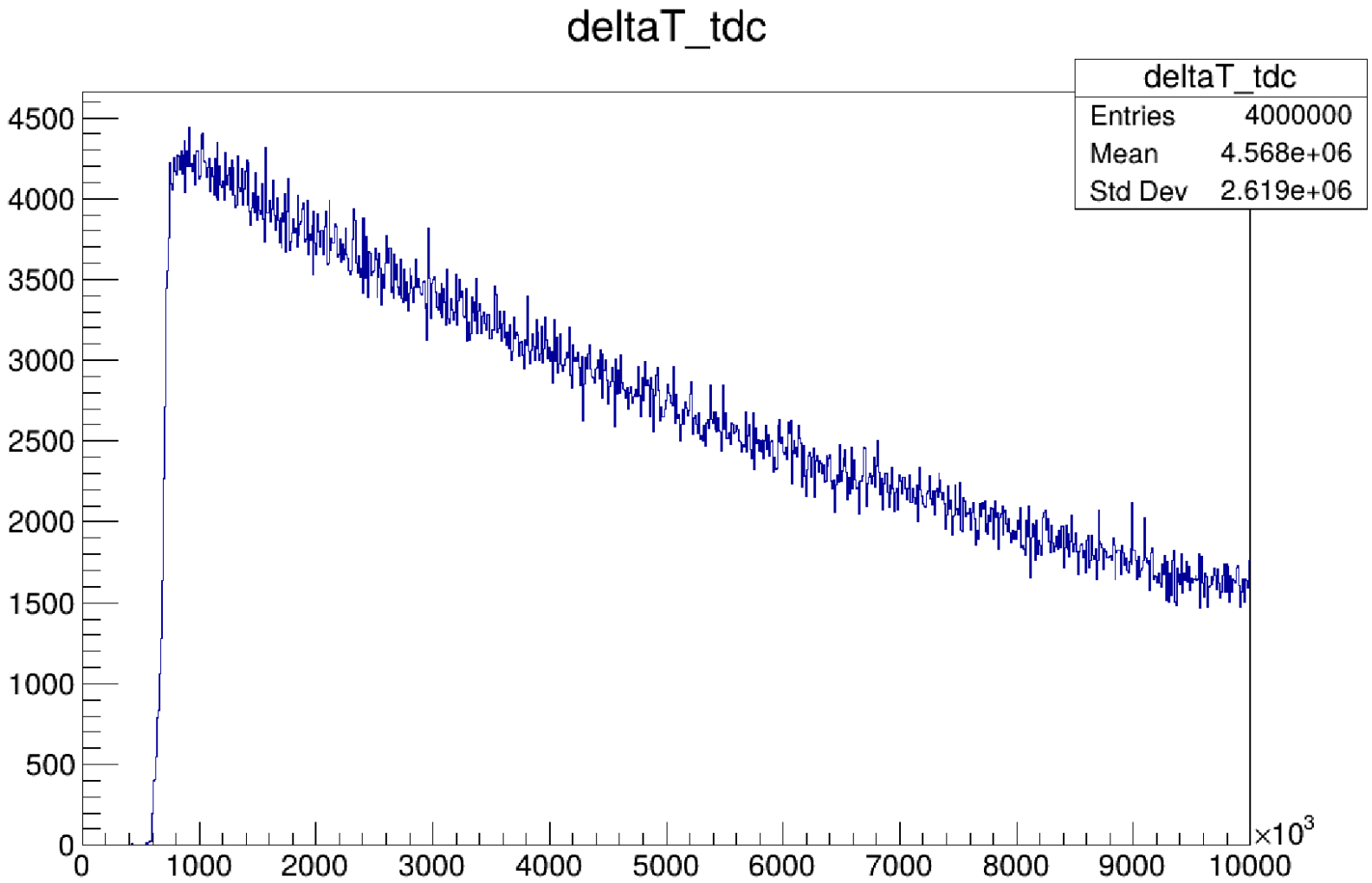
TDC time vs PPG time



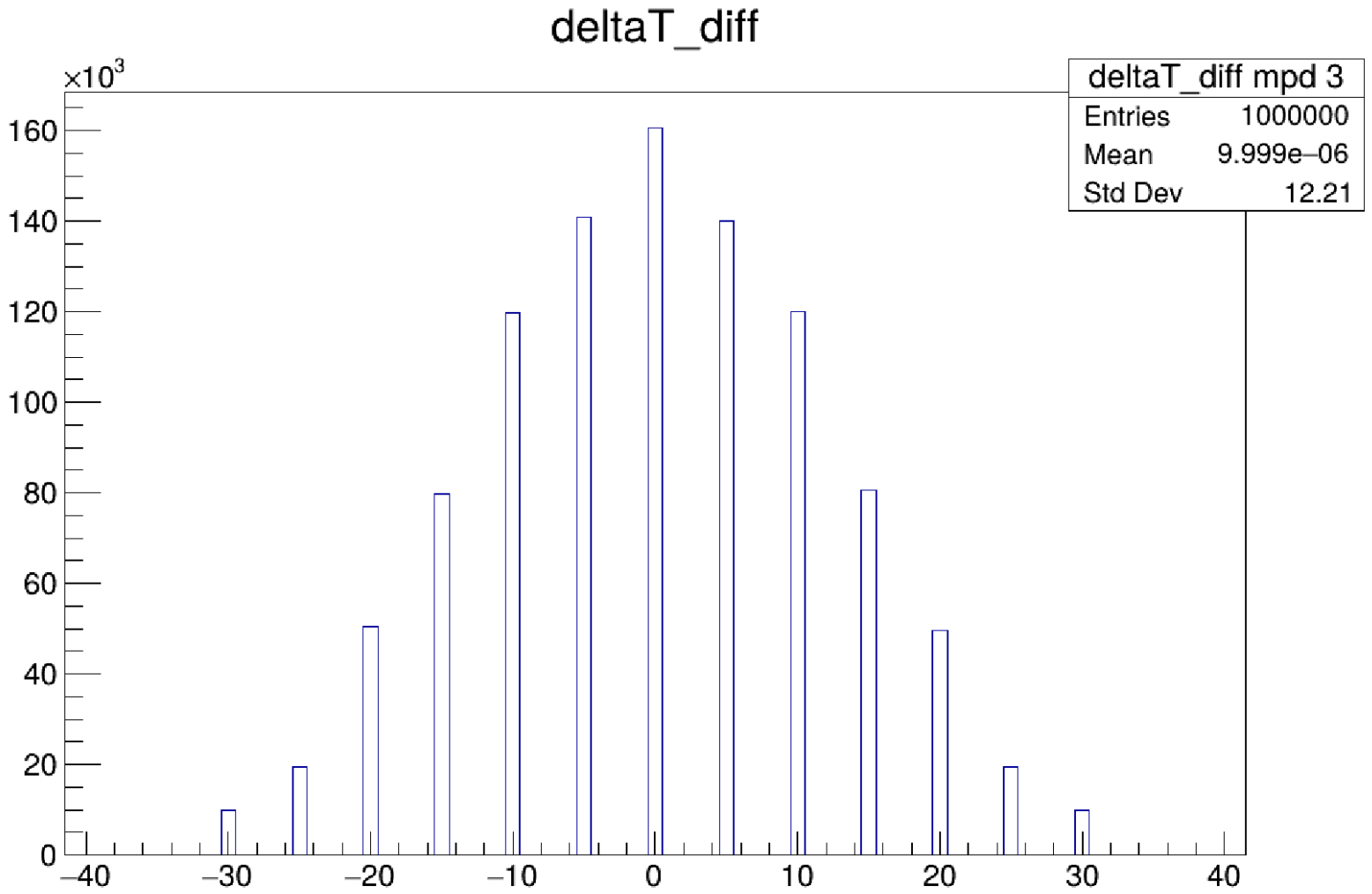
MPD Time Difference (MPD time – previous MPD time)



TDC Time Difference (TDC time – previous TDC time)

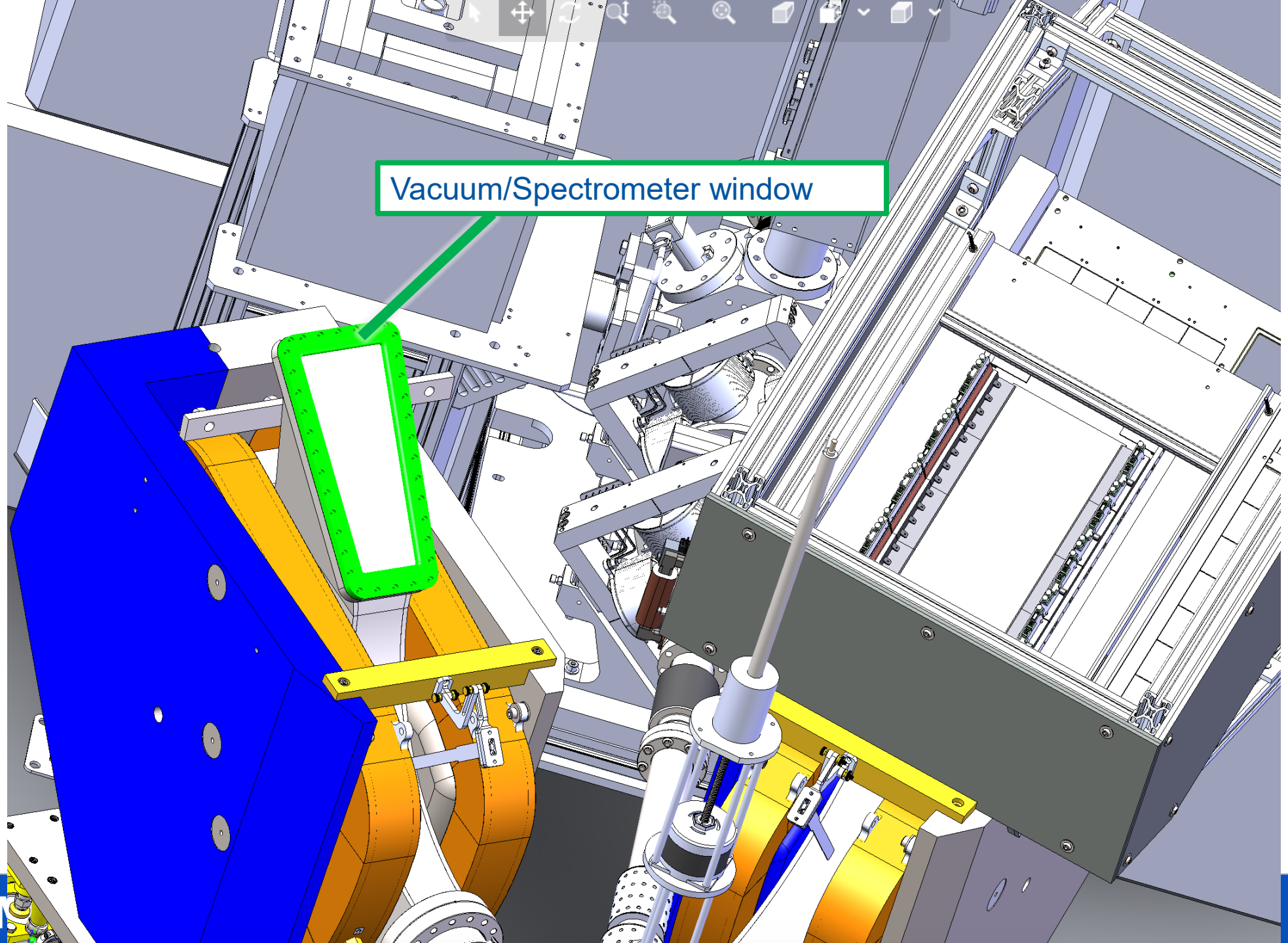


MPD Time Diff. – TDC Time Diff.



Updating GEM Geometry

- Will finish today
- Assuming minimal translation of GEMs to match survey data


Vacuum/Spectrometer window

Top surface of bottom GEM:

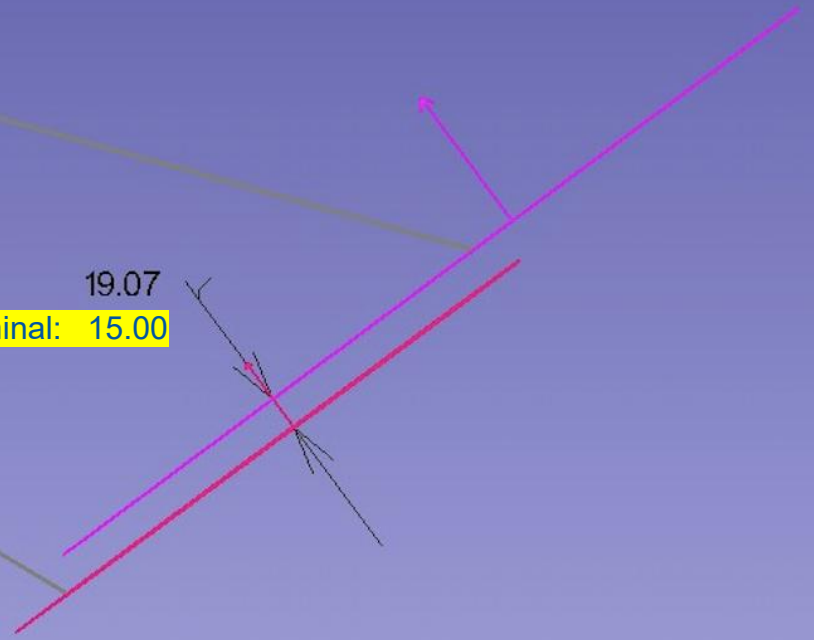
North Dipole Window Edge			
Proj.Ang	Rx from Y	Ry from Z	Rz from X
	89.9502	-36.6299	179.9330

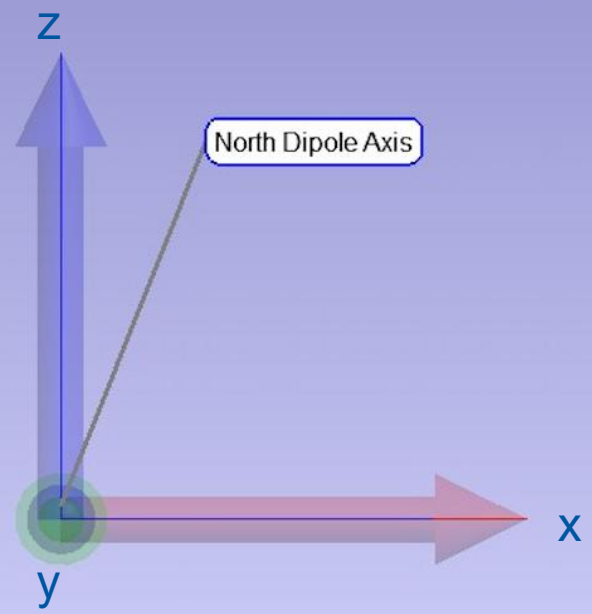
Nominal: 90 -36.54 -180

Top surface of spec window:

North Dipole Window Flange			
Proj.Ang	Rx from Y	Ry from Z	Rz from X
	90.5177	-36.4224	-179.2983

Nominal: 90 -36.54 -180





North DP Collimator Flange OD			
	X	Y	Z
Origin	410.17	0.41	0.19
Proj.Ang	Rx from Y	Ry from Z	Rz from X
	0.6233	89.9985	0.1381

North Dipole Fiducials As-Built::1	
X	799.96
Y	204.92
Z	270.03

North Dipole Fiducials As-Built::2	
X	712.06
Y	204.88
Z	87.92

North Dipole Fiducials As-Built::3	
X	639.93
Y	204.89
Z	0.08

Top surface of bottom GEM:

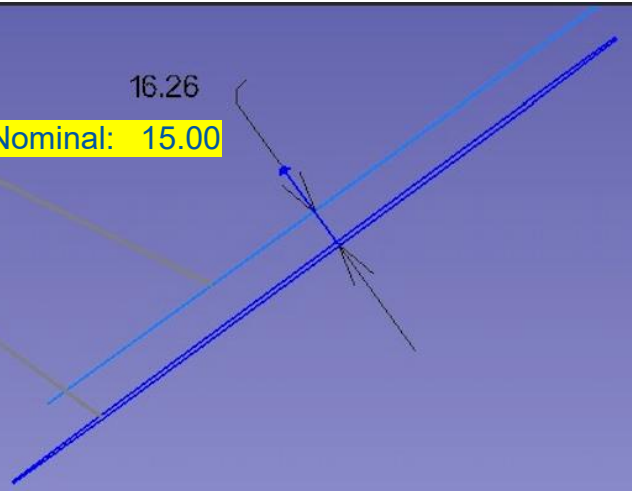
South Dipole detector top			
Proj.Ang	Rx from Y	Ry from Z	Rz from X
	90.6904	-35.8587	-179.0449

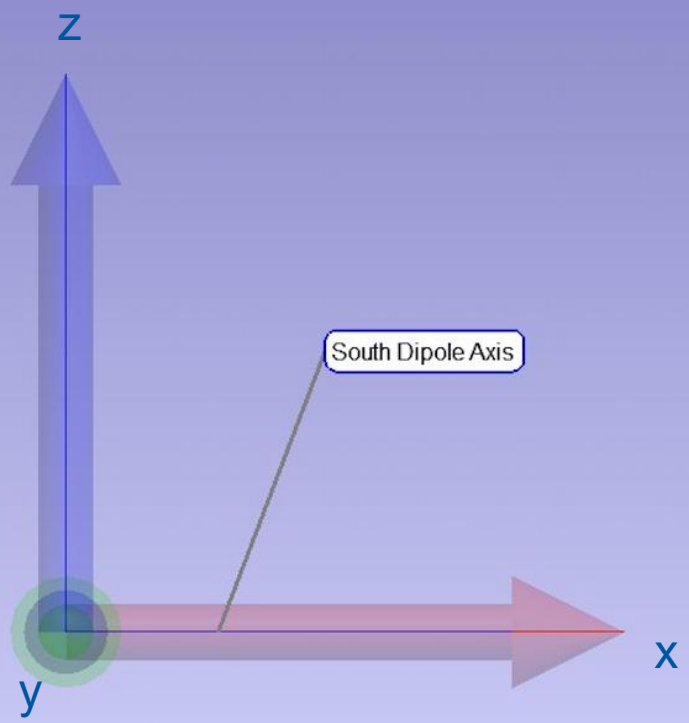
Nominal: 90 -36.54 -180

Top surface of spec window:

South Dipole Window Flange			
Proj.Ang	Rx from Y	Ry from Z	Rz from X
	90.9584	-36.2815	-178.6945

Nominal: 90 -36.54 -180





South DP Collimator Flange OD			
	X	Y	Z
Origin	414.75	-1.10	-0.82
Proj.Ang	Ry from Z	Rz from X	
	90.2171	0.7375	

South Dipole Fiducials Final Alignment::1	
X	800.10
Y	205.07
Z	269.98

South Dipole Fiducials Final Alignment::2	
X	712.23
Y	205.16
Z	87.83

South Dipole Fiducials Final Alignment::3	
X	640.09
Y	204.97
Z	-0.05


LEFT
Y
X
Z